4 A length of wire is held taut between two points M and P as shown in Fig. 4.1. A signal generator which produces an alternating current of variable frequency is passed through the wire and a pair of magnets is placed on either side of the wire.

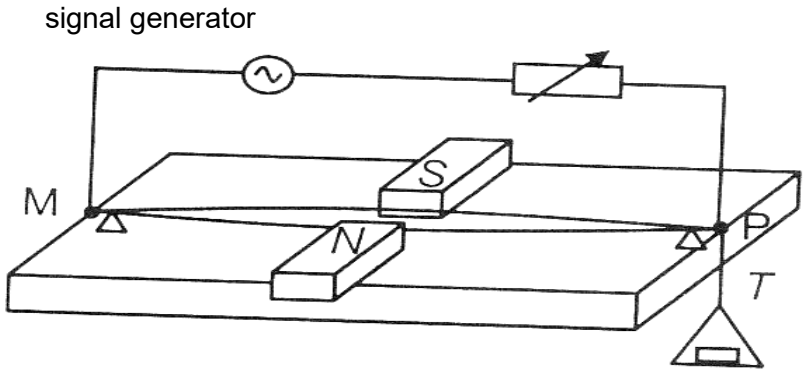

Fig. 4.1

The frequency of the alternating current is gradually increased from zero. A stationary wave is set up as shown in Fig. 4.1 when the frequency is 10 Hz.

(a) Explain how the stationary wave is formed on the wire when an alternating current is passed through it.

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…….. [4]

(b) The distance between M and P is 0.60 m.

Calculate the wavelength of the stationary wave formed.

wavelength = ……..m [1]

(c) Determine the speed of the wave.

speed = ……. m s^{-1} [2]

(d) Sketch the stationary wave formed in the space below when the frequency of the alternating current is adjusted to 30 Hz.

[2]